

is necessary measurement.

To use an accelerometer, we first need to fix the zero point of the accelerometer.

Steps to set the zero point for your accelerometer:

1. Set the variable `zero_point` in the below program to zero i.e. `zero_point = 0;`
2. Hold the board straight (parallel to the ground) and watch the serial monitor.
3. When the `xAcc` and `yAcc` values show same value, that value is the zero point
4. Change the value of variable `zero_point` in the program disregarding the sign of the value.
 - a. `xacc` and `yacc` will show negative value (-290 in our case). Disregard the negative sign and set `zero_point = 290` in the sketch.

An accelerometer is a very sensitive device and is used when fast measurements are necessary, but the device is also sensitive to external disturbances that change its value and therefore needs to be used with care.

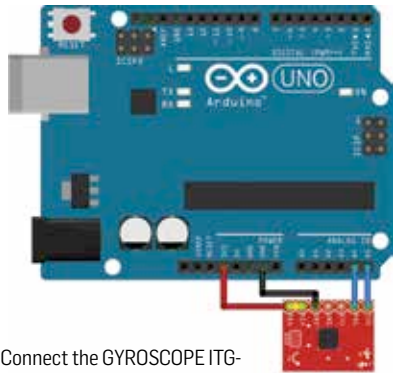
Using a Gyroscope

Programming a Gyroscope isn't as simple as programming the Accelerometer. It uses the I2C (Inter IC Connect) communication protocol via the SDA (data line) and SCL (clock line). You'll need to understand the data sheet of Gyroscope to understand the code. We recommend asking one of your friends who's an electronics expert to help you understand the code.

You can find a copy of the code at <http://dgit.in/ardgyrocode> and related explanation at <http://dgit.in/ardgyroexp>.



Accelerometer ADXL335 connected to the Arduino™ board. The Vcc voltage for the accelerometer ADXL335 in above figure is 3.3 V



Connect the GYROSCOPE ITG-3200 to the Arduino™